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**THE EFFICIENT CORRUPTION HYPOTHESIS
AND THE DYNAMICS BETWEEN
ECONOMIC FREEDOM, CORRUPTION, AND
NATIONAL INCOME**

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ABSTRACT

Corruption is an endemic problem in many developing economies. As a result, economists and political scientists have extensively studied corruption and its relationship to economic prosperity. An important mediating factor is the level of economic freedom. Determining causality is difficult as income, economic freedom, and corruption are arguably endogenous. We explicitly model this endogeneity using a panel Vector Auto Regression (pVAR) framework. The pVAR models we estimate are able to explicitly model this endogeneity better than the single-equation panel data models previously used in the literature. We have three primary data sources. Gross Domestic Product (GDP) per capita from the World Bank measures national income. Country-level corruption data comes from the World Bank's Control of Corruption database. The source of our country-level measure of economic freedom is the Fraser Institute's *Economic Freedom of the World* annual report. Our data set runs from 2002-2013 for over one hundred countries. We begin our results by finding that income, corruption, and economic freedom are integrated, but not cointegrated, thus a pVAR is appropriate. A one-time increase in corruption has a negligible effect on economic freedom and national income. An increase in economic freedom decreases corruption and increases GDP per capita. Increases in national income have no meaningful effect on corruption but have a small and positive effect on economic freedom. We then split our sample to see if there is a difference in results between high economic freedom countries and low economic freedom countries and do not find a difference. In summary, economic freedom decreases corruption and improves the economy and these results hold regardless of the level of economic freedom or whether corruption is absolute or relative. The policy implications from our research are straightforward. To the extent that resources to devote towards corruption are scarce, a focus on improving economic freedom not only leads to less corruption, but higher national income.

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INTRODUCTION

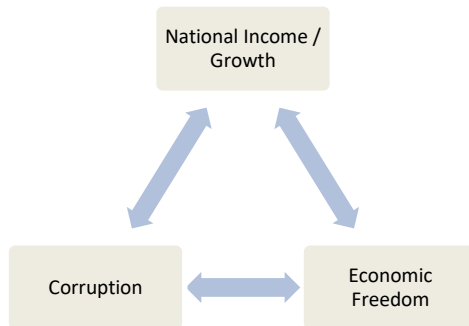
Economists and political scientists seem endlessly fascinated by the relationship between corruption and economic performance (Shleifer and Vishny, 1993; Mauro, 1995; Campos, 1999; Bologna, 2016). The earliest thinking was that a well-functioning economy relied on proper institutions.¹ Corrupt bureaucrats personified inefficient institutions. The bribes they demanded were seen as an unnecessary tax on businesses, which would invariably cause distortions, deadweight loss, inefficiency, and poverty.

That thinking was turned on its head in the 1960s, primarily by Leff (1964) and Huntington (1968), who argued that corruption – in some circumstances – provided a much-needed safety valve from stifling bureaucracy. In such circumstances, corruption “greased the wheels” of commerce by greasing the palms of corrupt officials. This line of thought was revived more recently in the work of Shleifer and Vishny (1993) and Mauro (1995). The common thread in most modern corruption research is that corruption may be beneficial—as a second best—when red tape and regulation is high.² Another way of putting it is that corruption might improve upon the state of affairs when there is little economic freedom in a country.³

Research on the relationship between corruption, national income/growth and economic freedom can be categorized as follows:

- (1) Does economic freedom or national income affect corruption?
- (2) Does corruption or national income affect economic freedom?
- (3) Does economic freedom or corruption affect national income?

FIGURE 1: THE RELATIONSHIP BETWEEN CORRUPTION, ECONOMIC FREEDOM, AND GROWTH



Taken together, it seems as though everyone acknowledges that freedom, corruption, and income are jointly determined as depicted in Figure 1. And yet, to our knowledge no one has investigated *all* of the endogenous connections between them. We fill this gap by using a panel VAR framework to answer all of these questions simultaneously.

LITERATURE REVIEW

Why are some countries so riddled with corruption? Do corrupt actors have so much to gain with so little to lose? Does a poor economy provide fertile ground for corruption? Or does corruption flourish according to the institutional setting in which it is planted? Does corruption find a place to hide, and indeed thrive, among the overgrown weeds of bureaucracy? Does corruption plant its own seeds, so that it is self-perpetuating?

There is a large empirical and theoretical literature on corruption trying to answer these questions. Banerjee et al. (2013) provide a good overview of this literature. For our purposes, we are most interested in the literature on how economic freedom (institutions) affect corruption. Saha et al. (2009) find that economic freedom and democracy generally provide enough sunshine to combat corruption. However, democracy may increase corruption if a country is currently highly regulated. Pieroni and d'Agostino (2013) investigated whether the institutional environment, specifically economic freedom, is inversely correlated with corruption. They find that "market competition may be bad for corruption when institutions are weak, as they often are in developing countries although, in general, the aggregate index confirms that competition reduces corruption... A lack of government regulations may actually yield more corruption in less developed countries, whereas standard recipes for greater freedom can be applied in developed countries" (Pieroni and d'Agostino, p. 70).

A number of other empirical papers also explore economic freedom's relationship with corruption. Goel and Nelson (2005) consider whether political or economic freedom (or its components) affect corruption, concluding that economic freedom (especially in the monetary realm) is most effective at combating corruption. Graeff and Mehlkop (2003) examine the effect of economic freedom and its individual components on corruption. They find that the magnitude of economic freedom's effect on corruption depends upon whether the country is rich or poor. Shen and Williamson (2005) treat EFW as exogenous and find that it decreases corruption, however, they do not consider the alternative causal scenario where corruption might affect economic freedom. Using panel data and the Arellano and Bond (1991) method to instrument for corruption and investment, Swaleheen and Stansel (2007) find that corruption is bad for economic growth when economic freedom is low, but positive when economic freedom levels are high. Heckelman and Powell (2010) find the opposite of Swaleheen and Stansel (2007), namely that corruption positively affects growth in low economic freedom environments but erodes as economic freedom increases. Carden and Verdon (2010) find that some types of corruption can lead to increases in economic growth when economic freedom is low.

Corruption is but one possible determinant of economic freedom. The question of why some countries are more economically free than others has been a growing area of research in economics. One of the earliest papers in this area is Dawson (1998), who finds a positive relationship between political freedom and economic freedom. Recent papers using more years of data find that political freedom or democratic transitions lead to economic freedom (Aixalá and Fabro, 2009; Rode and Coll, 2012). Hall (2016), however, finds no evidence that democracy has led to institutional convergence. March et al (2017) find that higher initial GNI per capita is associated with larger increases in economic freedom.

Regulations that lower economic freedom clearly benefit individuals who attempt to seek rents through corruption. Kurer (1993) developed a theoretical model where the degree of red tape is endogenous to a model of corruption. That is, corrupt public servants will often increase the amount of red tape and inefficiency, so that they can then provide citizens with a convenient escape route—for a fee. Corruption, in this case, causes a decline in economic freedom, and this decline further reduces income over and above the usual costs of rent seeking. Similarly, Kaufmann and Wei (1999) developed a formal model where bureaucracy is endogenously determined with the size of the bribe. The prospect of ill-gotten gains invites the bureaucrat to create additional red tape, so that they might sell a way around the rules. Again, corruption diminishes economic freedom.

By far the largest literature on corruption relates to how it effects national income or growth. For example, corruption can be bad for growth because of its effect on investment (Mauro, 1995). Meon and Sekkat (2005) find that corruption directly lowers growth in addition to indirectly lowering growth by diminishing investment. Several studies have found that corruption decreases foreign direct investment (Cuervo-Cazurra, 2006; and Egger and Winner, 2005; Brada et al, 2012). Pavlik (2018) highlights the role that uncertainty surrounding corruption plays with respect to growth.⁴

Murphy et al. (1991) also argue that the prospect of being a wealthy corrupt bureaucrat might entice young entrepreneurial talent from productive to extractive pursuits. Corruption might also exacerbate cash and credit constraints among budding entrepreneurs, thereby misallocating technology and capital (Murphy et al., 1993). In a similar vein, Choi and Thum (2004) present a theoretical model where the threat of corruption changes the incentives facing entrepreneurs to make lasting investments. Instead, they opt for the flexibility of fly-by-night investments. Such investments do not provide for sustained economic growth.

Kaufmann and Wei (1999) develop a formal model where the amount of red tape is endogenously determined with the size of the bribe. Firms find themselves in a prisoner's dilemma where bribery is individually rational, yet costly. Endogeneity effectively eliminates the "grease the wheels" hypothesis in their model. Further, they find that manufacturing firms who bribe have higher costs than those who do not bribe. Kaufmann and Wei are strongly in the camp that corruption is inefficient. Fisman and Svensson (2007) provide additional microeconomic evidence at the firm level that suggests increases in bribery payments are correlated with decreased rates of firm growth. Extrapolating to the country level corruption is clearly sand in the wheels of commerce.

In a closely related paper, Mo (2001) considers the transmission mechanism through which corruption might affect economic growth. Corruption can affect various factors, which in turn, affect growth. He finds evidence that corruption's effect on political instability has the most impact on growth, followed by the share of private investment, and human capital. However, Mo does not consider whether the effect of corruption might depend upon the institutional setting of the country. While he controls for political rights, he does not control for economic institutions. Further, by including it additively, his model does not allow institutions to mitigate the effects of corruption. Finally, he does not exploit the panel-nature of the available data.

Controlling for economic and political institutions is important given the work of de Vaal and Ebben (2011). They develop a theoretical model where the effect of bureaucratic corruption on growth depends upon economic institutions. Corruption in an

inefficient bureaucracy increases efficiency. In an otherwise healthy institutional setting, however, it is an economic bad. According to de Vaal and Ebben (2011, p. 12), “Only when the amount of political stability or property rights protection is above some threshold value, corruption affects these institutions negatively, depressing economic growth. This implies that the institutional framework is important for determining the corruption-growth relationship....”

In all of the literature, the endogeneity of corruption is a widely recognized problem, especially for the efficient corruption hypothesis. In his review of the literature, Bardhan (1997, p. 1323) explains that

“It is usually presumed that a given set of [regulatory] distortions are mitigated or circumvented by the effects of corruption; but quite often these distortions and corruption are caused or at least preserved or aggravated by the same common factors. The distortions are not exogenous to the system and are instead often part of the built-in corrupt practices of [the]...system.”

Svensson (2005, pp.36-7) clearly articulates this point with respect to the efficient corruption hypothesis:

“The proponents of ‘efficient corruption’... take the distortions circumvented by the corrupt actions as given. In most cases, distortions and corruption are caused by, or are symptoms of, the same set of underlying factors. As Myrdal (1968) pointed out, corrupt officials may not circumvent distortions, but instead actually cause greater administrative delays to attract more bribes.”

For this reason, we build toward a fully articulated model where economic institutions measured by economic freedom, corruption, and the economy all interact to determine each other.

DATA AND A FIRST LOOK

To measure corruption, we employ two different measures of corruption from the World Bank’s Control of Corruption dataset. (1) *CE*, an absolute index of corruption, and (2) *CP*, a country’s corruption percentile (i.e. relative ranking). *CE* gives the country’s corruption score in units of a standard normal distribution, i.e. ranging from approximately -2.59 to 2.06. That is, it is in terms of standard deviations.⁵ Being a percentile, *CP* ranges from 0 to 100. While there are other corruption datasets, such as those by Political Risk Service (PRS), the International Country Risk Guide (ICRG), and Transparency International’s Corruption Perceptions Index, these measures of corruption are all highly correlated with each other. This is not surprising given that *CE* is comprised of data from many sources, including PRS and ICRG. Given the high correlation across measures and the public availability of the World Bank data, we opt to use *CE* and *CP* in our analysis.

We also employ the World Bank’s national income data (*rGDPpc*), measured as real GDP per capita in constant 2005 US dollars. Data on economic freedom (*EFW*) are

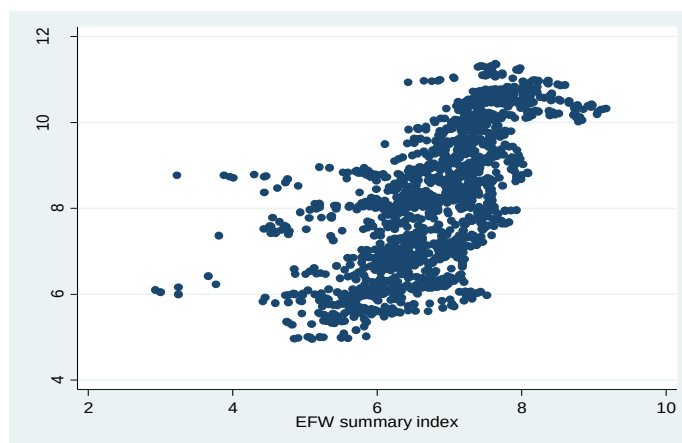
taken from the Fraser Institute's Economic Freedom of the World dataset (Gwartney et al. 2016). The EFW index measures the extent to which the policies and institutions of a country are consistent with economic freedom.⁶ Chain-linked and unadjusted data were available.⁷ We used the unadjusted numbers from 2002-2013.⁸ EFW is reported on a 0-10 scale, with larger values representing higher levels of economic freedom. While there are currently 159 countries in the EFW report, we need at least nine annual observations for each country using our empirical approach. Thus we exclude from our data set any country that was not in the EFW annually beginning in 2005. The data are summarized in Table 1.

TABLE 1: SUMMARY STATISTICS

Variable	Obs	Mean	St. Dev.	Min	Max
rGDPpc	1,755	11,440	15,976	135	86,129
EFW	1,589	6.8	0.9	2.9	9.2
CE	1,767	-0.06	1.03	-2.55	1.82
CP	1,767	48.82	28.71	0.00	100.00

To get a better sense of the relationship between corruption, economic freedom, and national income we created scatterplots. The first thing one sees when looking scatterplots of corruption, economic freedom, and national income is how strongly correlated the variables are. Figure 1 presents a scatterplot of (the log of) real per capita national income and economic freedom (EFW). The two variables are strongly positively correlated: countries with more economic freedom and less governmental interference tend to have healthier richer economies.

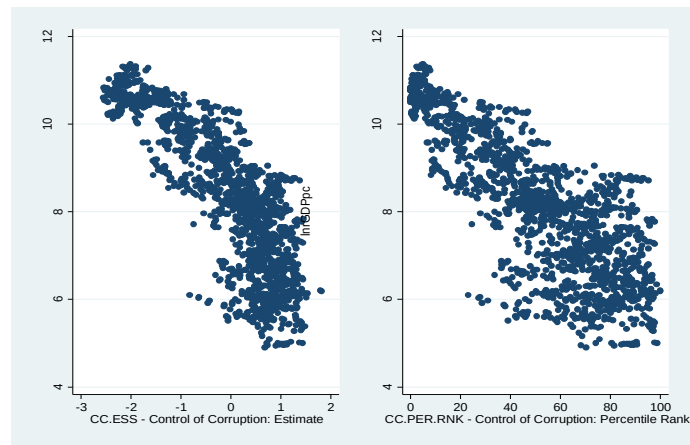
FIGURE 1: ECONOMIC FREEDOM AND NATIONAL INCOME



The negative correlation between national income and corruption is on display in Figure 2. There, we present two different measures of corruption: *CE* measures a country's index of corruption in absolute terms; *CP* measures a country's corruption, relative to other

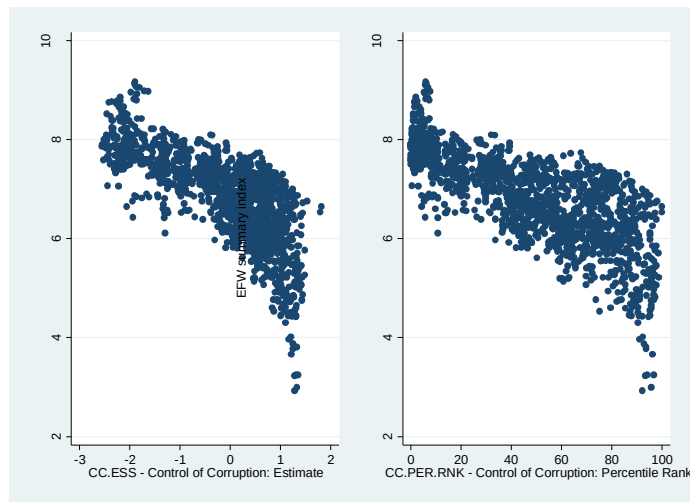
countries. Regardless of how it is measured, countries that are more corrupt tend to be much poorer.

FIGURE 2: CORRUPTION AND NATIONAL INCOME



Finally, Figure 3 illustrates the correlation between corruption and economic freedom. The two variables are negatively related. That is, countries that are freer economically tend to be less corrupt.

FIGURE 3: ECONOMIC FREEDOM AND CORRUPTION



A MORE DETAILED LOOK: PANEL VECTOR AUTOREGRESSIONS

One of the largest obstacles to resolving the question of causality is that all three variables—income, economic freedom and corruption—are likely endogenous. Corruption decreases incomes; low incomes make it more likely that people engage in corruption. Economic freedom changes the returns to corruption; bureaucrats may decrease economic freedom to further exploit gains from corruption. Economic freedom directly increases national income; poor countries may require increased regulation because they have poorly developed markets (a dubious hypothesis, but a logically possible one). The causal arrows cannot be established using theory alone. These questions require empirical analysis.

We explore fully the reciprocal dynamics between economic freedom, corruption, and national income via a panel VAR model. A VAR model grows exponentially more complicated with each additional variable and lag that is added. Rather than reporting all the coefficients, we show the impulse response functions (IRFs) from each model. Before we can estimate a VAR, regardless of whether it is in panels or not, we must first verify that the data are stationary (integrated). Second, if they are integrated, we must verify that they are not cointegrated. If they are integrated, but not cointegrated, then a VAR model can be estimated in the first-differences.

For each country in the dataset ($N=136$), and for each variable involved ($\ln rGDPpc$, EFW , CE , and CP), we conducted an Augmented Dickey Fuller test of stationarity. ADF tests have been found to be sensitive to the number of lags included in the model. To circumvent this difficulty, we estimated the ADF tests for lags of 1, 2, and 3.⁹ We tested at the 5% level. Simple Type-I error would be expected to lead to $N=7$ false positives ($136 \times 0.05 = 6.8$). In each instance, the number of countries that were found to be integrated far exceeded the number that would be expected from Type-I error. This was true, regardless of whether the model included a deterministic trend or not. For example, 16 (21) countries' GDPs were found to be integrated, when tested against a 3-lag model with (without) a deterministic trend.

Having established that the variables are integrated, we must check whether they are cointegrated. A panel cointegration test was not computable given the unbalanced dataset and the small number of time periods in the dataset.¹⁰ Thus, we conduct a series of country-by-country tests. We calculated a set of Engle Granger (1987) tests for cointegration between $\ln rGDPpc$, EFW , and corruption (CE or CP). Only two countries showed any cointegration when corruption was measured by CE ; only one, when corruption was measured by CP . Similar results held in a test including trend in the model: 4 (CE) and 3 (CP) countries showed cointegration. These numbers are far fewer than the 7 (6.8) countries that would be expected to show cointegration from Type-I error. Thus, we conclude that income, corruption, and economic freedom are not cointegrated. Thus, we do not need to estimate a Vector Error Correction Model (VECM), but rather a VAR.

The literature on VARs is fractured regarding estimating a VAR on integrated variables. Christopher Sims (1980) implies that differencing is not necessary and would throw away important information. Others, such as Peter Phillips (1998) disagree. He argues that IRFs from unrestricted VARs are inconsistent in the presence of unit roots. Given these problems of spurious regression, we focus on pVARs with differenced data.

How many lags should be included in the differenced pVAR models? We calculated the coefficient of determination for pVARs in differences with lags of one

through four. Regardless of whether corruption was measured as *CE* or *CE*, the coefficient of determination indicated that models with three lags were preferred.

In Figure 4, we show the IRFs from a pVAR in differences, where corruption is measured by *CE*, the absolute score on corruption. Figure 5 reports same for *CP*, the percentile rank of corruption. The two sets of IRFs show similar dynamics, only differing by their magnitudes. That is, the overall dynamics do not seem to depend on whether corruption is measured in absolute terms, or relative to the corruption of other countries. Both sets of estimates have eigenvalues inside the unit circle, indicating stability.

FIGURE 4: PVAR DLNRGDPPC DEFW DCE, LAGS(3)

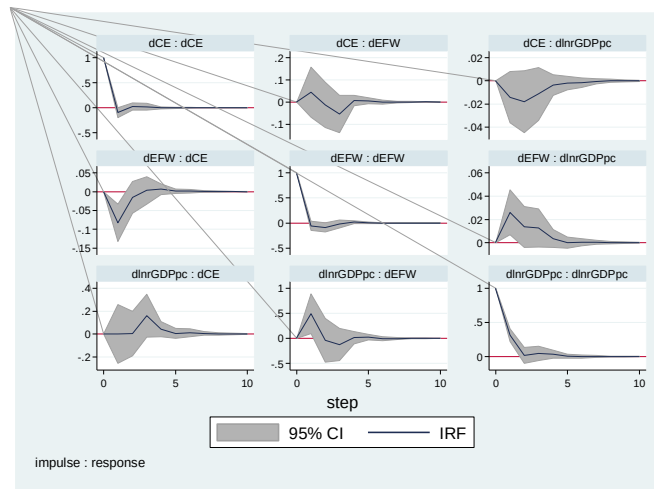
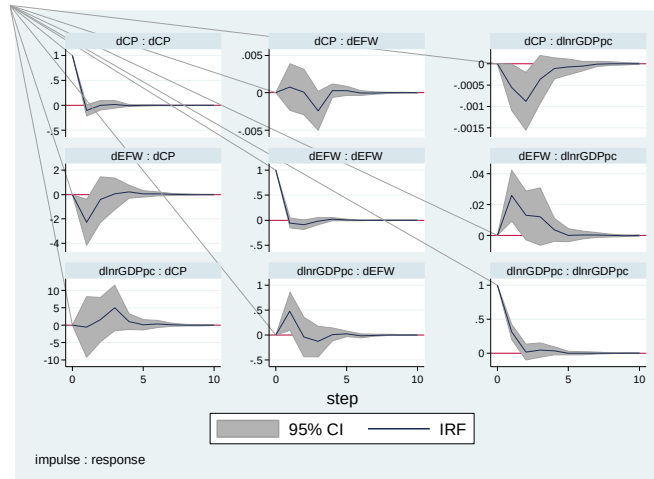


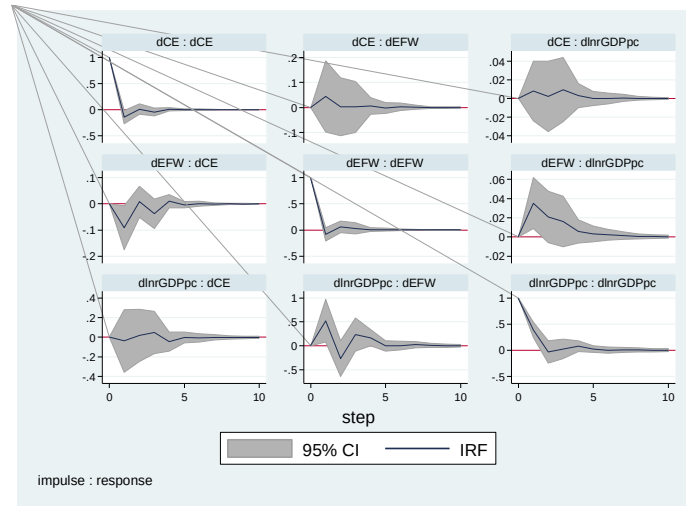
FIGURE 5. PVAR DLNRGDPPC DEFW DCP, LAGS(3)



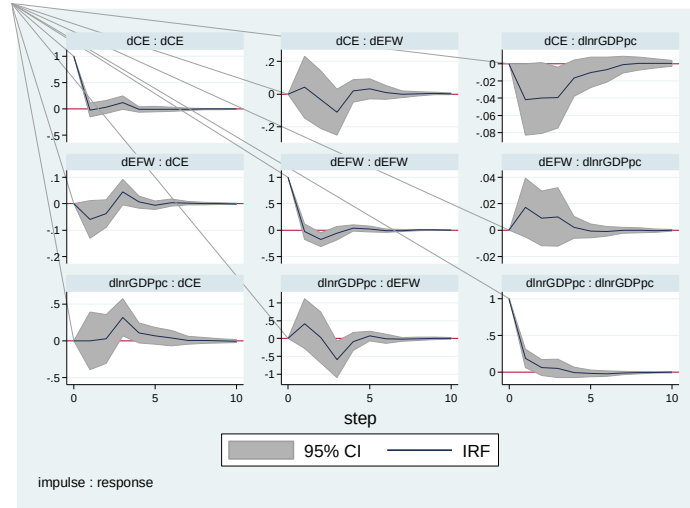
Figures 4 and 5 show that an increase in (the change of) corruption ($d.CE$ or $d.CP$) is not necessarily lingering: it tends to diminish over time. Moreover, this one-time increase in corruption has a negligible effect on economic freedom and GDP.¹¹ An increase in economic freedom, on the other hand, decreases corruption and increases per capita wealth accumulation. Finally, increases in wealth accumulation have no meaningful effect on corruption. Increases in wealth accumulation, however, are correlated with small yet statistically significant increases in economic freedom. In summary, economic freedom decreases corruption and improves the economy and these results hold regardless of whether corruption is absolute or relative.

Does the effect of corruption depend on whether the country is already over-regulated in terms of economic freedom? Are the proponents of the efficient corruption hypothesis such as Leff (1964), Huntington (1968), Shleifer and Vishny (1993) and Mauro (1995) correct? Is corruption a second-best escape-valve from over-burdensome restrictions on free trade? To answer these questions, we split our sample into two groups: countries that are more free than average ($EFW > 6.8$), and those that are less ($EFW < 6.8$)? If the efficient corruption hypothesis is correct, then increases in corruption will decrease incomes in high-EFW countries, but will decrease incomes by less (or even raise them) in low-EFW countries.

FIGURE 6A: PVAR DLNRGDPPC DEFW DCE IF EFW $\geq$ 6.802, LAGS(3)



To limit the proliferation of figures, we restrict ourselves to estimating the model corruption measured only as CE . Figure 6a shows the IRF from a pVAR on economically free countries; Figure 6b repeats this for economically unfree countries. For both samples, the estimates pass the eigenvalue test for stability, with all eigenvalues inside the unit circle. In Figure 6a we see, once again, the same familiar dynamics between the three variables. The fact that shocks to $d.CE$ have the same effect on $d.lnrGDPPc$, regardless of whether we are looking at economically free or un-free countries, would seem to imply that the efficient corruption hypothesis is merely wishful thinking.

FIGURE 6B. PVAR DLNRGDPPC DEFW DCE IF EFW $\leq$ 6.802, LAGS (3)

CONCLUSIONS

Income, economic freedom, and corruption interact in complex ways. All three variables are arguably endogenous. In this paper, we explicitly modeled the endogeneity using a panel VAR framework. The pVAR models we estimate are better able to model endogeneity than the single-equation panel data models that have been estimated elsewhere in the literature. That is, we are better able to capture the reciprocal causality between the variables. The pVAR models that we estimate provide evidence that corruption and regulations have a negative effect on national income.

Many previous studies in the literature acknowledge that the institutional setting helps determine income, but few examine the possibility that the setting determines the impact of corruption on income. The two exceptions are Swaleen and Stansel (2007) and Heckelman and Powell (2010). Our final set of pVARs detects no evidence that corruption provides a safety valve from stifling government regulation. This is most likely because the prospect of ill-gotten gains creates the regulation in the first place. That is, the regulation is endogenous. The pVARs here reveal that, once this endogeneity is taken into account—that is, once we allow for the possibility that regulation increases corruption—we find that corruption makes countries poorer.

That said, it is hard to prove a negative. Panel VARs require estimating many coefficients, which decreases the available degrees of freedom, and makes detecting statistically small changes difficult.¹² This is doubly true when the sample is split in half to test on high-EFW and low-EFW subsamples. Perhaps the effect exists, but is not detectable with the current amount of data. Once more data are collected, we might be able to confirm the efficient corruption hypothesis. But at this point, we argue that there is not enough evidence to support this hypothesis.

One hypothesis worth considering for future research is that weak economies, which usually are already struggling with corruption, have little to no efficiency to spare.

In this case, a little bit of additional corruption would have a greater impact on those economies. This might be termed the “increasing marginal cost of corruption” hypothesis. This could be tested by splitting the sample into highly-corrupt, and negligibly-corrupt groups, and re-estimating the pVARs. If the hypothesis is correct, then a one unit increase in corruption would decrease incomes more in the highly-corrupt sample than in the negligibly corrupt countries. We leave this for future research. An additional research idea to explore with more years of data would be the relative importance of each area of economic freedom to corruption.

ENDNOTES

* “*In terms of economic growth, the only thing worse than a society with a rigid, over-centralized, dishonest bureaucracy is one with a rigid, over-centralized, honest bureaucracy*” (Huntington, 1968, p.386).

¹ There is a large empirical literature on the effect of institutions following Knack and Keefer’s (1995) seminal paper. Unfortunately, this literature is too large for us to do justice to it here.

² Dawson and Seater (2013) demonstrate the negative effect of regulations on growth.

³ Economic freedom is certainly multi-dimensional and more than just bureaucracy (Bologna and Hall, 2014; Gwartney et al. 2016). However, the majority of economic freedom areas (property rights, rule of law, freedom to trade, regulations) are highly correlated (Beaulier et al. 2016).

⁴ There is also a large literature on the relationship between economic freedom and growth. See, Compton et al. (2011) for a representative paper and Hall and Lawson (2014) for a summary of this literature.

⁵ The Control of Corruption dataset is part of the World Bank’s *Worldwide Governance Indicators*. According to their documentation, “Control of corruption captures perceptions of the extent to which public power is exercised for private gain, including both petty and grand forms of corruption, as well as ‘capture’ of the state by elites and private interests (Worldwide Governance Indicators, 2017).” As an index, their data is compiled from a number of sources. Interested readers should see the full documentation available on the WGI site.

⁶ For more background on the EFW index and the literature that has used it, see Hall and Lawson (2014).

⁷ There exists a second measure of economic freedom, the *Index of Economic Freedom* (Miller et al. 2019). Although the two economic freedom indices differ in substantive ways, the correlation between the two indices over our sample is 0.861. In empirical results that could not be included due to space limitations, but which are available upon request, we re-do our analysis here using the *Index of Economic Freedom* and find very similar results.

⁸ A handful of countries are not in the data set in 2002. For example, Armenia’s first year in the EFW is 2004. We have data on 118 countries starting annually in 2002 and by 2005 we have a total of 136 countries in the data set.

⁹ We estimated 3,288 ADF tests: 137 (the number of countries) x 4 (variables) x 2 (with and without trend) x 3 (the number of lags).

¹⁰ As noted earlier, our data spans from 2002-2013 for 118 countries and a minimum of 2005-2013 for the remaining 17 countries in our unbalanced panel.

¹¹ This is one place where the unreported *Index of Economic Freedom* results differ. Using the *Index of Economic Freedom*, we find that increasing corruption decreases economic freedom.

¹² It is for this reason it is not possible to look at all areas of economic freedom and determine the area that would be most consistent with the efficient corruption hypothesis.

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